

UNIT 3: ENERGY CHANGES & RATES OF REACTION

Knowledge	Thinking/Inquiry	Communication	Application
/20	18 /18	10 /10	10 /10

SHORT ANSWER QUESTIONS

Answer the following questions in the space provided.

THINKING/INQUIRY (8 MARKS)

1. In a coffee-cup calorimeter, 60.0 mL of 0.700 mol/L sodium hydroxide solution was neutralized with 40.0 mL of excess sulfuric acid solution. The initial temperature of the mixed solutions was 20.0°C and rose to 25.6°C. Assume the density of the solution is 1.0 g/mL and the heat capacity of the solution is 4.18 J/g°C. (4 marks) Find molar enthalpy (per mol of NaOH) in kJ/mol

$$\Delta T = 25.6^\circ\text{C} - 20.0^\circ\text{C} = +5.6^\circ\text{C}$$

$$m_{\text{NaOH}} = 60.0 \text{ mL} \cdot \frac{1 \text{ g}}{1 \text{ mL}} = 60.0 \text{ g}$$

$$m_{\text{sur}} = 60.0 \text{ mL} + 40.0 \text{ mL}$$

$$= 100.0 \text{ mL} \cdot \frac{1 \text{ g}}{1 \text{ mL}}$$

$$= 100.0 \text{ g}$$

$$\text{Molarity NaOH} = 0.700 \text{ mol/L} \cdot \frac{1 \text{ L}}{1000 \text{ L}} = 0.000700 \text{ mol/mL}$$

$$0.000700 \text{ mol/mL} \cdot 60.0 \text{ mL} = 0.0420 \text{ mol NaOH}$$

$$\Delta H = -m_{\text{sur}} c \Delta T$$

$$= - (100.0 \text{ g}) (4.18 \text{ J/g}^\circ\text{C}) (5.6^\circ\text{C})$$

$$\Delta H = -2340.8 \text{ J} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = -2.3408 \text{ kJ}$$

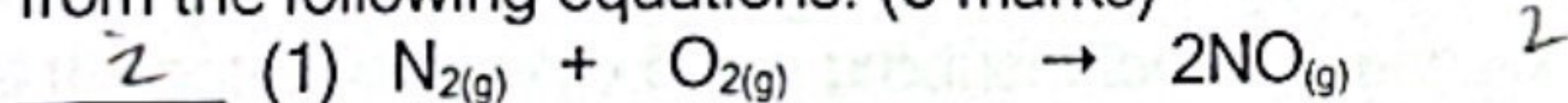
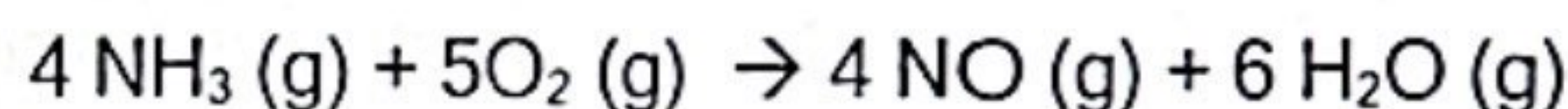
$$\Delta H_{\text{f, mol}} = \frac{\Delta H}{n_{\text{NaOH}}} = \frac{-2.3408 \text{ kJ}}{0.0420 \text{ mol}}$$

$$= -55.73 \text{ kJ/mol}$$

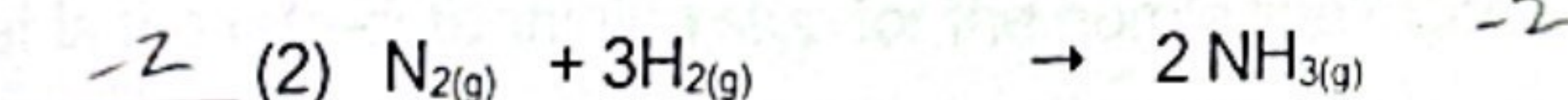
SDZ.

$$= -55.73 \text{ kJ/mol}$$

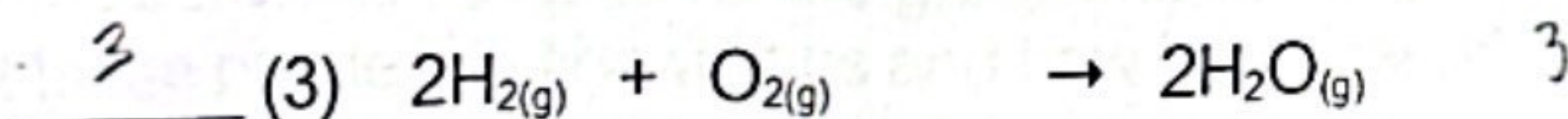
2. Calculate the enthalpy change for the following reaction from the following equations: (5 marks)



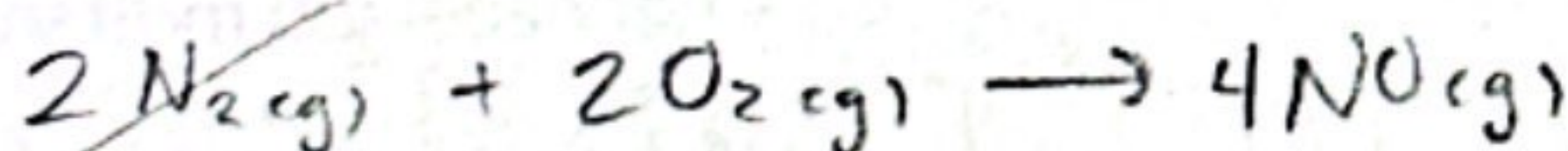
$$\Delta H_1 = -180.5 \text{ kJ}$$



$$\Delta H_2 = -91.8 \text{ kJ}$$



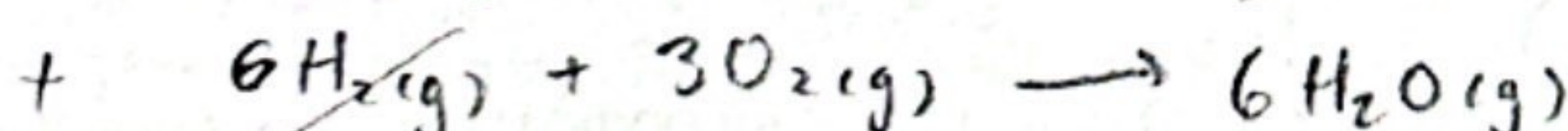
$$\Delta H_3 = -483.6 \text{ kJ}$$



$$\Delta H_1 = (2)(-180.5 \text{ kJ}) = -361 \text{ kJ}$$

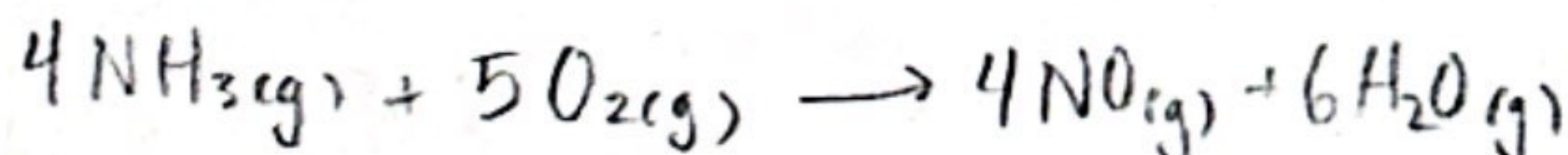


$$\Delta H_2 = -2(-91.8 \text{ kJ}) = 183.6 \text{ kJ}$$

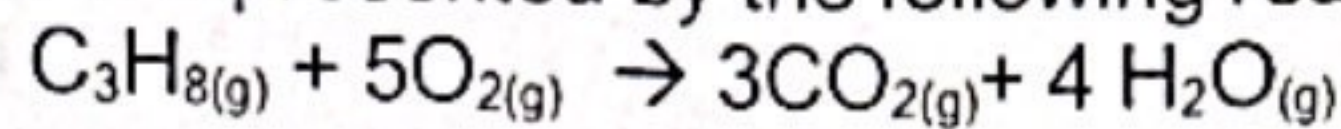


$$+ \Delta H_3 = 3(-483.6 \text{ kJ}) = -1450.8 \text{ kJ}$$

$$\Delta H = -1628.2 \text{ kJ}$$



3. The combustion of propane gas is represented by the following reaction: (2 marks)



(a) Explain whether the reaction would occur in a single step or in a series of steps. (1 mark)

A series since it's ^{very} hard to get all these reactants to collide at once (max 3 at once)

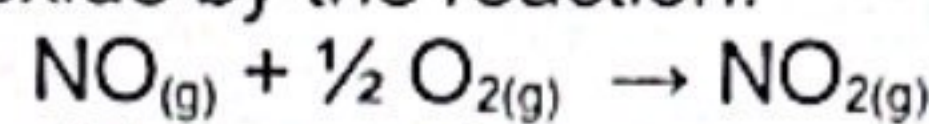
(b) If the rate of production of carbon dioxide gas is $3.5 \times 10^{-2} \text{ mol/L}\cdot\text{s}$, write the expressions and numerical values for the rate of reaction of oxygen gas. (2 marks)

$$\frac{1}{3 \text{ mol}} \left(\frac{\Delta [\text{CO}_{2(g)}]}{\Delta t} \right) = \frac{1}{5 \text{ mol}} \left(- \frac{\Delta [\text{O}_{2(g)}]}{\Delta t} \right)$$

$$\frac{5 \text{ mol}}{3 \text{ mol}} \left(\frac{\Delta [\text{CO}_{2(g)}]}{\Delta t} \right) = - \frac{\Delta [\text{O}_{2(g)}]}{\Delta t}$$

$$\frac{5 \text{ mol}}{3 \text{ mol}} (3.5 \times 10^{-2} \text{ mol/L}\cdot\text{s}) = 5.8 \times 10^{-2} \text{ mol/L}\cdot\text{s}$$

4. The formation of small amounts of nitric oxide in automobile engines is the first step in the formation of smog. Nitric oxide is readily oxidized to nitrogen dioxide by the reaction:



The following data were collected in a study of the rate of the rxn.

Trial	[O ₂] (mol/L)	[NO] (mol/L)	Initial rate (mol/(L·s))
1	0.0010	0.0010	7.10
2	0.0040	0.0010	28.4
3	0.0040	0.0030	255.6

(a) Use this information to determine the order of the reactants. Show your work! (2 marks)

for [O₂]: trial 1 → 2: as [O₂] × 4, R × 4 → 4¹ m = 1
 $0.0040 \div 0.0010 = 4$ $28.4 \div 7.10 = 4$

for [NO]: trial 2 → 3: as [NO] × 3, R × 9 → 3² n = 2
 $0.0030 \div 0.0010 = 3$ $255.6 \div 28.4 = 9$

(b) Write the rate law expression. (1 mark)

$$R = k [\text{O}_2]^1 [\text{NO}]^2$$

$$R = 7.1 \times 10^9 [\text{O}_2] [\text{NO}]^2$$

(c) What is the value of the rate constant, k? (2 marks)

Trial 1 values: $R = k [\text{O}_2]^1 [\text{NO}]^2$

$$k = R / [\text{O}_2]^1 [\text{NO}]^2$$

$$= 7.10 / (0.0010)(0.0010)^2$$

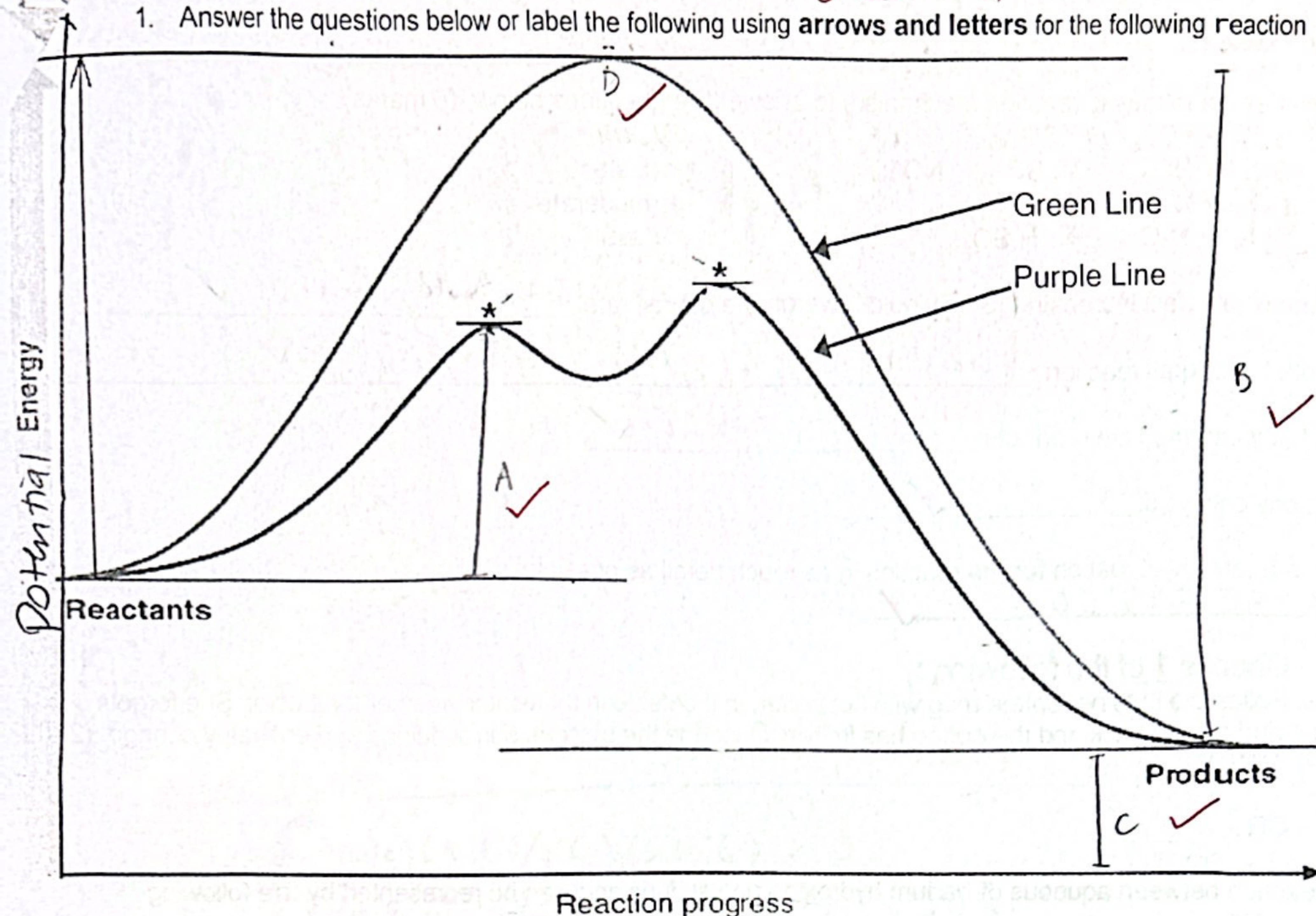
$$k = 7.1 \times 10^9$$

(d) What is the overall order for the reaction? (1 mark) $1 + 2 = 3$

COMMUNICATION (10 MARKS)

10 excellent

1. Answer the questions below or label the following using arrows and letters for the following reaction

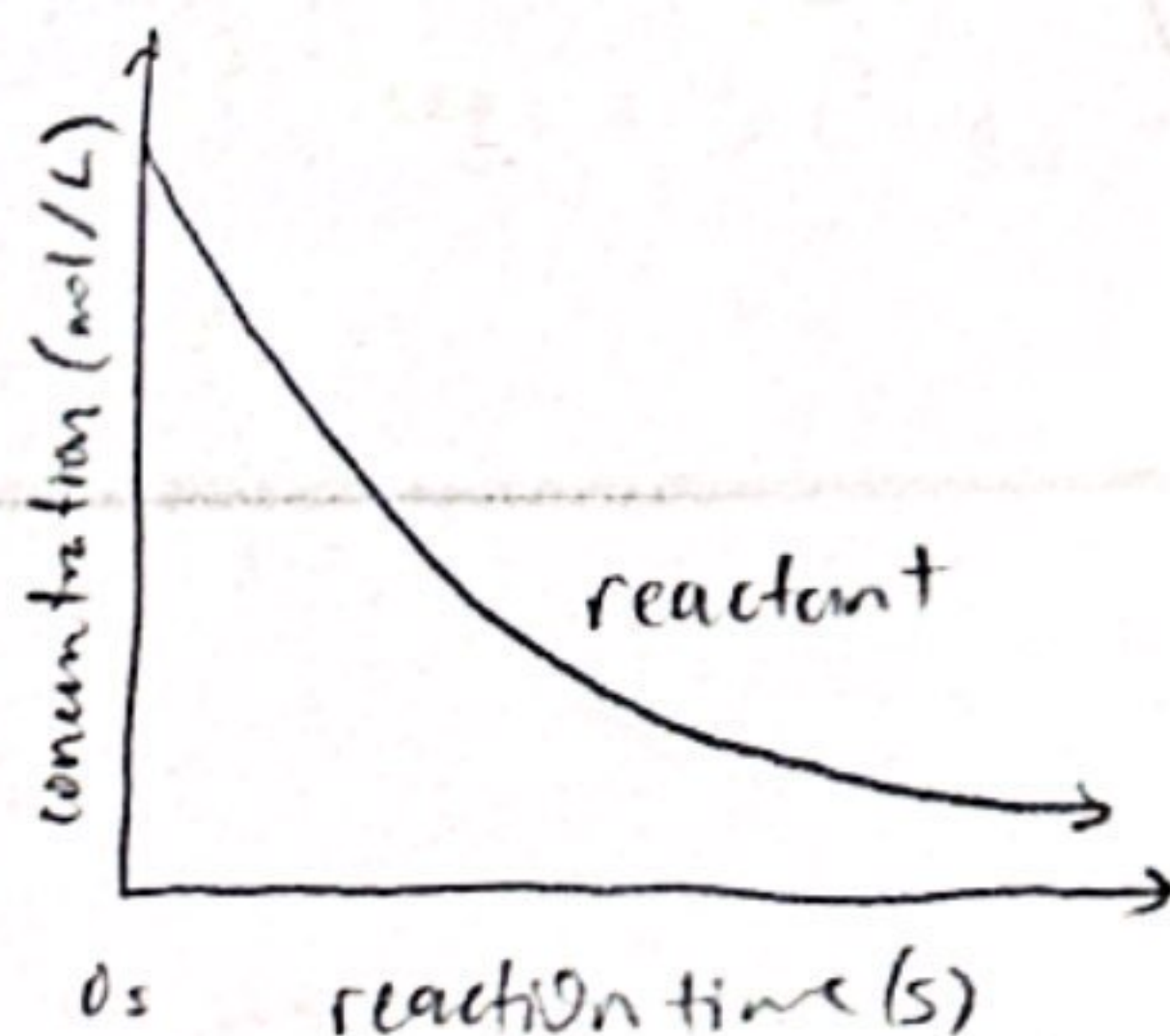


- Is this reaction exothermic or endothermic? (1 mark) exothermic
- Label the activation energy of the first elementary step of the purple line with letter A. (1 mark)
- Label the activation energy for the reverse reaction for the green line with letter B. (1 mark)
- Label the potential energy of the products for the green line with letter C. (1 mark)
- What is the rate-determining step for the purple line? Circle one: STEP 1 or STEP 2 (1 mark)
- Label the activated complex for the green line with the letter D. (1 mark)

g) What is the purple line showing us and how is it different from the green line? (2 marks so list two points- use vocabulary from our lessons.)

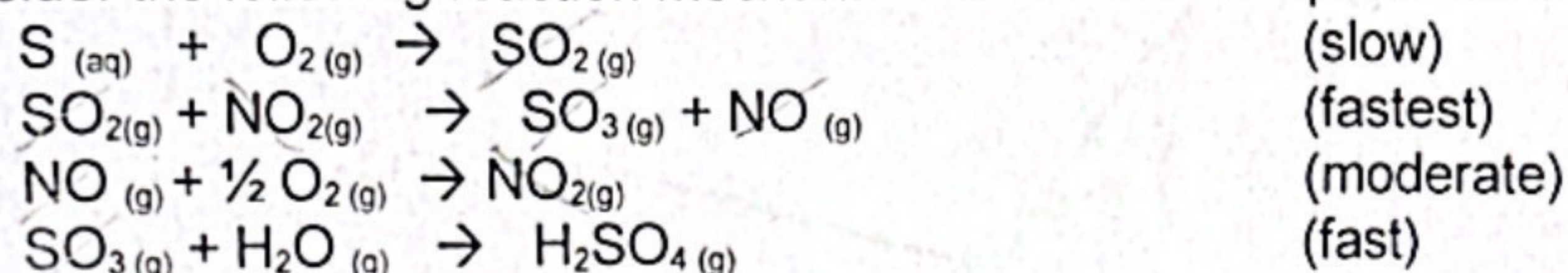
- the purple line is showing us the reaction of the green line if the reaction was given an easier pathway (lower activation energy threshold) through the use of a catalyst.
- It is different from the green line because rather than one step, the purple line goes through 2 elemental step reactions with lower required activation energy.

2. Sketch a concentration vs. time graph to represent the consumption of a reactant over the course of a reaction (2 marks)



APPLICATION (10 MARKS)

1. Consider the following reaction mechanism to answer the questions below. (8 marks)



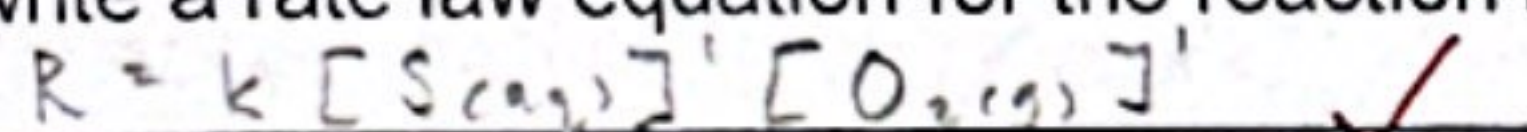
a) State what effect increasing $[\text{S}_{(\text{aq})}]$ would have on the overall rate: it would increase ✓

b) Write the overall reaction: $\text{S}_{(\text{aq})} + \frac{3}{2} \text{O}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{g})} \rightarrow \text{H}_2\text{SO}_{4(\text{g})}$ ✓

c) List any intermediate products: $\text{SO}_{2(\text{g})}$, $\text{SO}_{3(\text{g})}$, $\text{NO}_{(\text{g})}$ ✓

d) List any catalysts: $\text{NO}_{2(\text{g})}$ ✓

e) Write a rate law equation for the reaction in as much detail as possible:

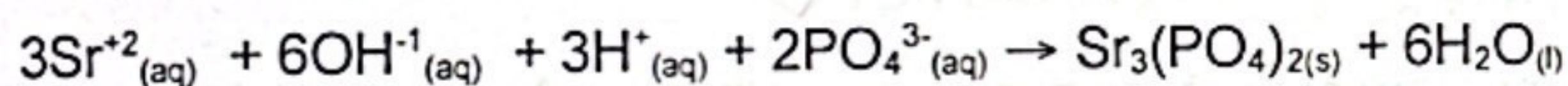


Choose 1 of the following :

2) Mrs. Policarpio puts her coffee mug with hot coffee in it outside in the winter weather for it cool. She forgets about it and comes back and the coffee has frozen. Describe the system, surroundings and enthalpy change (2 marks)

OR

The reaction between aqueous of barium hydroxide and sulfuric acid can be represented by the following equation:



List and describe 1 way that this reaction's rate could be measured. Give an explanation as to how it can be done. (2 marks)

System: coffee $\text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_2\text{O}_{(\text{s})}$ ✓

Surroundings: cup, cold winter air

The enthalpy change would be negative because the system loses energy overall, making the reaction exothermic.